Worksheet 8-2 B Particle Motion #2
AP Calculus Name_____

1.) A particle's velocity is given by $v(t) = t^2 + 2t - 15$, where t is measured in minutes, v is measured in meters per minute, and $s(t)$ represents the particle's position.

(a) If $s(1) = -3$, what is the value of $s(3)$?

(b) What is the net change in distance (displacement) over the first 5 minutes?

(c) What is the total distance traveled by the particle during the first 5 minutes? Show the set up AND your work.

2015 AB3—NO CALCULATOR

t (minute)	0	12	20	24	40
$v(t)$ (meters per minute)	0	200	240	–220	150

2.) Johanna jogs along a straight path. For $0 \le t \le 40$, Johanna's velocity is given by a differentiable function $v(t)$. Selected values of $v(t)$, where t is measured in minutes and $v(t)$ is measured in meters per minute, are given in the table above.

(a) Use the data in the table to estimate the value of $v'(16)$

(b) Using correct units, explain the meaning of the definite integral $\int_0^{40} |v(t)|dt$ in the context of the problem. Approximate the value of $\int_0^{40} |v(t)|dt$ using a right Riemann sum with the four subintervals indicated in the table.

(c) Bob is riding his bicycle along the same path. For $0 \le t \le 10$, Bob's velocity is modeled by $B(t) = t^3 - 6t^2 + 300$, where t is measured in minutes and $B(t)$ is measured in meters per minute. Find Bob's acceleration at time $t = 5$.

(d) Based on the model B from part (c), find Bob's average velocity during the interval $0 \le t \le 10$.

2017 AB5—NO CALCULATOR

3.) Two particles move along the x-axis. For $0 \le t \le 8$, the position of particle P at time t is given by $x_P(t) = \ln(t^2 - 2t + 10)$, while the velocity of particle Q at time t is given by $v_Q(t) = t^2 - 8t + 15$. Particle Q is at position $x = 5$ at time $t = 0$.

(a) For $0 \le t \le 8$, when is particle P moving to the left?

(b) For $0 \le t \le 8$, find all times t during which the two particles travel in the same direction.

(c) Find the acceleration of particle Q at time $t = 2$. Is the speed of particle Q increasing, decreasing, or neither at time $t = 2$? Explain your reasoning.

(d) Find the position of particle Q the first time it changes direction.

2016 AB2—A CALCULATOR MAY BE REQUIRED FOR THIS PROBLEM

4.) For $t \geq 0$, a particle moves along the x–axis. The velocity of the particle at time t is given by $v(t) = 1 + 2\sin\left(\frac{t^2}{2}\right)$. The particle is at position $x = 2$ at time $t = 4$.

(a) At time $t = 4$, is the particle speeding up or slowing down?

(b) Find all times t in the interval $0 < t < 3$ when the particle changes direction. Justify your answer.

(c) Find the position of the particle at time $t = 0$.

(d) Find the total distance the particle travels from time $t = 0$ to time $t = 3$.